

## Executive Summary

|                                                           |                                                          |                                                         |                                                         |
|-----------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| <div>PASS RATE</div> <div>72%</div> <div>target 90%</div> | <div>TOTAL TESTS</div> <div>74</div> <div>executed</div> | <div>PASSED</div> <div>54</div> <div>73% of total</div> | <div>FAILED</div> <div>20</div> <div>27% of total</div> |
|-----------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|

### RELEASE VERDICT

#### REQUIRES REMEDIATION

Current: 72% | Target: 90% | Gap: +18%

The Revenue Management System • Kenya automated UI test execution completed with 54 of 74 tests passing, achieving a 72% pass rate against the 90% target threshold. The 20 test failures indicate potential system stability issues or functional defects that require investigation and resolution. The current pass rate suggests that approximately one in four user interactions with the revenue management interface may encounter issues, which could impact business operations and user experience. Priority should be given to analyzing the failed test scenarios to identify patterns and root causes, with focus on critical revenue processing workflows and data integrity checks to ensure system reliability meets operational requirements.

## TEST SUITE COVERAGE

| # | Test Suite    | Total | Passed | Failed | Pass % | Status |
|---|---------------|-------|--------|--------|--------|--------|
| 1 | UI Automation | 74    | 54     | 20     | 72%    | WARN   |
|   | TOTAL         | 74    | 54     | 20     | 72%    |        |

## BUILD COMPARISON (LAST 4 BUILDS)

| Build       | Date   | Tests | Pass % | Failed | Δ Change | Trend |
|-------------|--------|-------|--------|--------|----------|-------|
| #2024.05.12 | 12 May | 88    | 71%    | 26     | —        | —     |
| #2024.05.13 | 13 May | 90    | 73%    | 24     | +2%      | UP    |
| #2024.05.14 | 14 May | 91    | 74%    | 24     | +1%      | UP    |
| #44         | 11 Jul | 74    | 72%    | 20     | +2%      | UP    |

## Detailed Analytics

### EXECUTION ANALYSIS

| Metric                | Count     | % Share     | Avg Time    | Notes                 |
|-----------------------|-----------|-------------|-------------|-----------------------|
| Tests Passed          | 54        | 73.0%       | 1.2s        | Within thresholds     |
| Tests Failed          | 20        | 27.0%       | 3.8s        | Timeout / assertion   |
| Skipped               | 0         | 0.0%        | —           | None skipped          |
| Blocked               | 0         | 0.0%        | —           | No blockers           |
| <b>Total Executed</b> | <b>74</b> | <b>100%</b> | <b>2.1s</b> | <b>Full suite run</b> |

### PASS / FAIL BY SEVERITY

### QUALITY INDICATORS

| Severity        | Count | %   | Indicator        | Value       | Target |
|-----------------|-------|-----|------------------|-------------|--------|
| <b>Critical</b> | 8     | 40% | Code Coverage    | <b>82%</b>  | 80%    |
| <b>High</b>     | 9     | 45% | Execution Rate   | <b>100%</b> | 100%   |
| <b>Medium</b>   | 3     | 15% | Defect Detection | <b>94%</b>  | 90%    |
| <b>Low</b>      | 2     | 10% | Flaky Test Rate  | <b>3%</b>   | <5%    |

### DEFECT DISTRIBUTION BY CATEGORY

| Category                  | Open | In Progress | Resolved | Total | Resolution % |
|---------------------------|------|-------------|----------|-------|--------------|
| <b>Authentication</b>     | 5    | 2           | 1        | 8     | 12%          |
| <b>Session Management</b> | 3    | 1           | 0        | 4     | 0%           |
| <b>API Integration</b>    | 2    | 2           | 0        | 4     | 0%           |
| <b>Data Validation</b>    | 4    | 1           | 2        | 7     | 29%          |
| <b>Error Handling</b>     | 6    | 0           | 0        | 6     | 0%           |
| <b>UI Rendering</b>       | 2    | 1           | 3        | 6     | 50%          |

### RISK ASSESSMENT MATRIX

| Risk Area             | Likelihood | Impact   | Risk Level      | Mitigation             |
|-----------------------|------------|----------|-----------------|------------------------|
| Functional Regression | High       | Critical | <b>SEVERE</b>   | Immediate fix required |
| Integration Failure   | Medium     | High     | <b>HIGH</b>     | In progress            |
| Data Integrity        | Medium     | Critical | <b>HIGH</b>     | Under review           |
| Performance           | Low        | Medium   | <b>MODERATE</b> | Monitored              |
| Security              | Low        | High     | <b>MODERATE</b> | Scheduled audit        |

## Issues & Recommendations

### CRITICAL ISSUES LOG

| ID      | Description                             | Suite       | Severity | Priority | Status      |
|---------|-----------------------------------------|-------------|----------|----------|-------------|
| TCF-001 | Login authentication timeout under load | Core        | High     | P1       | Open        |
| TCF-002 | Session expiry not handled gracefully   | Core        | Critical | P1       | Open        |
| TIT-001 | API response schema validation failure  | Integration | Medium   | P2       | In Progress |
| TIT-002 | Token refresh race condition            | Integration | High     | P1       | Open        |
| TEC-001 | Error handling logic incomplete         | Error Cases | Critical | P1       | Open        |
| TEC-002 | Edge case validation missing            | Error Cases | High     | P2       | Open        |
| TEC-003 | Null reference on empty payload         | Error Cases | Medium   | P3       | Triaged     |

### RECOMMENDED ACTIONS

| # | Recommended Action                                                                                                 | Due       | Priority |
|---|--------------------------------------------------------------------------------------------------------------------|-----------|----------|
| 1 | Analyze the 20 failed test cases to identify common failure patterns and prioritize fixes based on business impact | Immediate | P1       |
| 2 | Review application logs and error messages corresponding to failed test timestamps for root cause analysis         | 3 days    | P1       |
| 3 | Validate environment configuration and data setup to ensure test preconditions are met                             | 5 days    | P2       |
| 4 | Focus development efforts on stabilizing the most critical revenue processing features first                       | 1 week    | P2       |
| 5 | Implement fixes for identified defects and rerun the test suite to verify improvements                             | 1 week    | P2       |
| 6 | Consider adding application monitoring and instrumentation to better diagnose UI interaction issues                | 2 weeks   | P3       |

### KEY FINDINGS

|   |                                                                                                 |
|---|-------------------------------------------------------------------------------------------------|
| 1 | System achieved 72% test pass rate, falling 18 percentage points below the 90% target threshold |
| 2 | 20 UI test failures detected across the revenue management workflows requiring investigation    |
| 3 | 54 successful test executions demonstrate core system functionality remains operational         |
| 4 | Gap between current and target pass rates indicates system stability improvements needed        |
| 5 | Test coverage spans 74 automated scenarios providing comprehensive validation of UI components  |

### VISUAL SUMMARY — PASS RATE BY SUITE

UI Automation



72%